

Bulk Canvas → Moodle Migration

A step-by-step guide for administrators

Migrating many courses off Canvas at once with the **Canvas Uplifter** (`tool_canvasuplifter`) and **Automate** (`tool_automate`) Moodle admin tools.

Applies to	Canvas Uplifter 0.43.0+ and Automate 1.3.0+, on Moodle 5.0–5.2
Audience	Site administrators and managers running a Canvas migration
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1. Overview

Canvas Uplifter imports a single Canvas Common Cartridge (.imsc) export into Moodle: it can **analyse** a package (report what it contains and how cleanly each part maps to Moodle) or **build** a brand-new Moodle course from it. For a handful of courses you can drive it by hand from its own upload page.

When you are moving **many** courses, the Automate plugin drives Canvas Uplifter in bulk. You feed it a list of Canvas backup URLs and/or a server directory of packages; Automate turns each package into one background Canvas Uplifter job. This guide covers that bulk workflow end to end.

The two ways each package can be handled

Build courses now	A new, hidden Moodle course is created from each package automatically. Best when you trust the packages and want courses ready to review.
Analyse for later	Each package is fetched and a conversion report produced, but no course is built. An administrator reviews each report and builds the course manually from Canvas Uplifter when ready.

Why background jobs?

Nothing is converted inside your browser request. Every package becomes a background adhoc task, so a directory or URL list of hundreds of courses is queued instantly and then worked through by cron instead of timing out the page. They run under Moodle's ordinary site-wide task scheduling — how many run at once follows your *Server > Tasks* worker configuration.

2. Before you start

- **Both plugins installed.** Canvas Uplifter provides the import engine; Automate provides the bulk driver. Automate detects Canvas Uplifter automatically — the bulk-import feature only appears when it is present.
- **Capabilities.** You need `tool/automate:manage` to run Automate, and `tool/canvasuplifter:use` for the import itself. Building courses additionally needs `moodle/course:create` in the target category.
- **Cron running.** The imports run as background tasks, so Moodle cron must be active (*Site administration > Server > Tasks > Task processing*).
- **Trusted packages only.** Imported HTML is stored and displayed as authored. Only import .imsc packages from sources you trust.

Security: server-side URL fetch (SSRF)

When you import from a download URL, the site fetches it server-side through Moodle's SSRF-aware cURL layer, governed by *Site administration > Security > HTTP security*. On Moodle 5.0+ these settings block loopback and private ranges by default. Keep those defaults; uploading the file directly avoids the fetch entirely.

3. Turn on bulk Canvas import

The feature ships behind an off-by-default kill-switch. Enable it once:

Step 1 — Open Automate's settings

Go to *Site administration > Plugins > Admin tools > Automate > Settings*.

Step 2 — Allow bulk Canvas import

Tick **Allow bulk Canvas import**. This is re-checked at both queue time and run time, so clearing it later stops new and pending jobs.

Step 3 — (Optional) set a source directory

If you will import from a folder of packages on the server, set **Bulk Canvas import source directory** to that folder (a staging area holding `.imsc` / `.zip` files). You can skip this if you only import from URLs.

4. Prepare your packages

Option A — A list of Canvas backup URLs

In Canvas, export each course as a Common Cartridge and collect the download URLs — one per line. Each is fetched in the background.

```
https://school.instructure.com/exports/course-101.imsc  
https://school.instructure.com/exports/course-102.imsc  
https://school.instructure.com/exports/course-103.imsc
```

Option B — A directory of packages

Copy the exported `.imsc` files into the source directory you configured in step 3. The picker searches server-side and type-to-filters, so it scales to hundreds of files.

You can use either source, or both together in one run.

5. Run a bulk import

Step 1 — Open the bulk import page

Go to *Plugins > Admin tools > Automate > Bulk Canvas import*.

Step 2 — Choose your sources

Paste your URL list into the text area, and/or pick packages from the server directory. Preview the selection before queueing.

Step 3 — Choose what happens to each package

- **Build courses now** — a new course is created from each package (needs course-create permission in the target category).
- **Analyse for later** — each package is analysed only, ready for you to review and build manually.

Step 4 — Set the target category and options

Pick the course category new courses land in. Optional build settings (such as building a runnable quiz from each standalone question bank, or combining consecutive pages into a book/lesson) carry through to each job.

Step 5 — Queue the jobs

Confirm to queue every package as a background job. Watch progress in *Site administration > Server > Tasks > Task logs* — the tasks appear as **Analyse Canvas package** and **Build course from Canvas package** (queued under **Bulk Canvas import**).

6. Review & build staged imports

Packages you queued with **Analyse for later** appear under *Automate > Staged Canvas imports* once analysed. From there you can:

- **Review a report and build the course** — open a completed analyse job to reach Canvas Uplifter's status page, read the conversion report, and use **Build this course** when you are happy.
- **Jump to a built course** — a completed build links straight to the course it created.
- **See storage used** — the top of the page shows how much space your stored packages occupy.
- **Reclaim space** — tick finished imports and **Delete selected** to free their stored packages.

Deleting is safe

Deleting a finished import frees its stored .imsc package **once no other job still references it** (an analyse job and the build made from it share one package, so it is freed only when the last of them is deleted) **but keeps any course a build already created** — it reclaims cache, it does not undo an import. Only finished (done or failed) imports can be deleted, so a delete never interrupts an in-flight conversion.

7. Driving it from the command line

The same bulk job can run from the CLI — handy for very large migrations or scripting. Run with `--help` for all options, or `--list` to inspect the sources without queueing.

```
# Build a course from every URL in a file, into category 5:
php admin/tool/automate/cli/import_canvas.php \
--urls=/data/urls.txt --category=5 --mode=build --execute

# Analyse a directory of packages for later review:
php admin/tool/automate/cli/import_canvas.php \
--source=/data/canvas --category=5 --mode=analyse --execute

# Inspect what would be queued, without queueing anything:
php admin/tool/automate/cli/import_canvas.php --urls=/data/urls.txt --list
```

8. Monitor & troubleshoot

Where to watch	<i>Site administration > Server > Tasks > Task logs</i> . Each import narrates what it did.
A job failed	Open the job's status page from <i>Staged Canvas imports</i> for the error; failed jobs can be deleted and re-queued.
Quiz has no questions	Canvas sometimes exports a quiz shell without its item bank. It builds as a hidden placeholder; re-export from Canvas with item banks included.
A URL will not fetch	Check <i>Security > HTTP security</i> — the host or port may be blocked. Upload the file directly to bypass the server-side fetch.
Feature missing	Confirm both plugins are installed and Allow bulk Canvas import is ticked. Bulk import needs Canvas Uplifter 0.40.0+; the storage counter and selective delete need 0.43.0+.

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